SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Error Analysis Task

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 3 – Error Analysis Task
Weightage:	35% of CIA	**Duration / Marks:**	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:	Slot A	**Date:**	10-10-2026

Code of Conduct

I _K.Madhulika,_192524190_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _Madhulika_____

Instructions

- Read each question carefully; answer all four questions (100 in total). Off-Class Assessment Tool.
- Each question gives a student's faulty work. Do not solve from scratch: identify every error (cite it by statement, component, iteration or step number), explain its root cause, correct it with justification and apply the corrected concept. An error merely carried forward counts once.
- Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.
- Marks follow the Error Analysis rubric: error identification, root cause analysis, correction and justification, application of concepts, and clarity and presentation.
- Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	**Q1**
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	**Q1**
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	**Q1**
4. The Nature of Environments	Task-environment properties and their classification	**Q1**
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	**Q1**
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	**Q2**
7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	**Q2, Q4**
8. Searching for Solutions	Search graph, frontier, explored set, node expansion	**Q3**

9. Uninformed Search Strategies	Error tracing of BFS, DFS and uniform-cost search	**Q3**
10. Search with Partial Information	Sensorless problems, belief states, percept update	**Q4**

Annexure A – Error Analysis Task

Question 1: Foundations of AI, Intelligent Agents and Rationality – Error Analysis of Concept Notes

Topic: Foundations and history of AI, agents and environments, rationality, nature of environments, structure of agents

A junior engineer prepared the revision notes below on the foundations of AI and intelligent agents. The notes contain errors; some statements may be correct. Analyse them statement by statement.

No.	Statement in the notes
S1	The Turing Test (1950) judges a machine to be intelligent if its internal reasoning always follows the laws of logic. It is therefore the standard example of the "thinking rationally" approach to AI.
S2	The "AI winters", periods of reduced funding and interest, occurred in the 1950s, before the 1956 Dartmouth workshop. That workshop then revived the field and coined the term "artificial intelligence".
S3	A rational agent always achieves the best possible outcome because it knows the actual result of every action in advance. Hence a taxi that picks a route later blocked by an unforeseeable landslide was not rational.
S4	PEAS description of an automated taxi – Performance measure: steering, accelerator, brake, horn. Environment: roads, other traffic, pedestrians, weather. Actuators: safe, fast, legal and comfortable trip. Sensors: cameras, GPS, speedometer.
S5	Chess played without a clock is a partially observable, stochastic, episodic, static and discrete task environment.
S6	A model-based reflex agent keeps an internal state, updated using knowledge of how the world evolves and how its own actions affect it. It can therefore act sensibly even when the current percept does not reveal the full state of the world.

(a) Identify which statements are correct and which are incorrect. For each incorrect statement, quote the erroneous phrase or claim (one statement contains more than one error).

Statement	Correct / Incorrect	Erroneous phrase / claim
S1		
S2		
S3		
S4		
S5		
S6		

(b) Root cause analysis: for every error, explain the underlying misconception, i.e. which concept has been misunderstood and why the statement goes wrong.

Error (statement / part)	Root cause (misconception)

(c) Rewrite each faulty statement correctly and justify your correction. For a statement that is already correct, write "No change" and justify why.

Statement	Corrected statement	Justification
S1		
S2		
S3		
S4		
S5		
S6		

(d) Apply the corrected concepts to the following case. A self-driving taxi chooses a highway route that every available traffic report showed to be clear; an unforeseeable landslide then blocks the road.

(i) Using the four factors that decide whether an agent is rational, argue whether the taxi's choice was rational.

(ii) Name the properties of the taxi's task environment that make perfect foresight impossible.

(iii) Describe one situation in which a simple reflex taxi would behave poorly, and state what a model-based reflex taxi keeps in order to avoid it.

(i)	
(ii)	
(iii)	

Question 2: Problem-Solving Agents and Problem Formulation – Error Analysis of a Formulation

Topic: Problem-solving agents, problem formulation

A student formulates the **Missionaries and Cannibals** problem for a problem-solving agent. Three missionaries and three cannibals stand on the left bank of a river, with a boat that carries one or two people and cannot cross empty. The boat is on the left bank at the start. Missionaries must never be outnumbered by cannibals on either bank (a bank with no missionaries is safe). The objective is to move everyone to the right bank using the **fewest crossings**.

Component	Student's formulation
State representation	(M, C) = number of missionaries and cannibals on the left bank
Initial state	(3, 3)
Actions	Take the boat across with 1M, 2M, 1C, 2C or 1M + 1C
Goal test	M = C
Legality check	A state is legal if C ≤ M on the left bank (or M = 0 there)
Path cost	Number of people carried in the crossing
Note N1	The agent works in the order: search → goal formulation → problem formulation → execution.
Note N2	The state space contains 16 states, of which 10 are legal.

(a) For each component and note, state whether it is correct or incorrect; for each incorrect one, state what is wrong.

Component / Note	Correct / Incorrect	What is wrong
State representation		
Initial state		
Actions		
Goal test		
Legality check		
Path cost		
Note N1		

Component / Note	Correct / Incorrect	What is wrong
Note N2		

(b) Root cause analysis: for every error, explain the underlying cause, i.e. which principle of problem formulation or of the problem-solving agent has been violated.

Error (component / note)	Root cause

(c) Give the corrected version of every faulty component or note and justify each correction.

Component / Note	Corrected formulation	Justification

(d) Apply your corrected formulation.

(i) Complete the table of boat loads available from the initial state, writing each resulting state as (M, C, boat side) and whether it is legal, with the reason.

Boat load	Resulting state (M, C, boat)	Legal? Reason
1M		
2M		
1C		
2C		
1M + 1C		

(ii) Give one state that the student's legality check accepts although it is actually illegal.

(iii) Using your corrected path cost, state the cost of an optimal solution.

(ii)	
(iii)	

Question 3: Searching for Solutions and Uninformed Search – Error Analysis of Search Traces

Topic: Searching for solutions, BFS, DFS, uniform-cost search

A student ran BFS, DFS and uniform-cost search (UCS) on the weighted undirected graph below. The start node is S and the goal node is G. **Conventions:** graph search with an explored set; neighbours expanded alphabetically; goal test when a node leaves the frontier; ties broken alphabetically. BFS and DFS ignore edge costs while searching, but path costs are measured with them. **BFS:** do not add a node that is already explored or already in the queue. **DFS:** use a stack with its top on the left; push children so that the alphabetically first child is popped first; do not push a node that is already explored or already on the stack. **UCS:** write entries as Node:g(parent) and replace an entry whenever a cheaper path to a frontier node is found.

Edge	Cost	Edge	Cost
S – C	3	B – G	3
S – D	3	D – E	2
A – B	2	D – G	6
A – C	6	E – G	3
B – D	4		

Student's BFS trace (queue initially [S])

Iteration	Node Dequeued	Queue after Expansion	Explored Set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[B, E, G, A]	{S, C, D}
4	B	[E, G, A]	{S, C, D, B}
5	E	[G, A]	{S, C, D, B, E}
6	G (goal)	[A]	{S, C, D, B, E}

BFS result: path S – D – G, cost 9, 6 nodes removed from the frontier.

Student's DFS trace (stack initially [S], top on the left)

Iteration	Node Popped	Stack after Pushing Children
1	S	[D, C]
2	D	[G, E, B, C]
3	G (goal)	[E, B, C]

DFS result: path S – D – G, cost 9, 3 nodes removed from the frontier.

Student's UCS trace

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), A:9(C), G:9(D)
5	B	7	A:9(C), G:9(D)
6	A	9	G:9(D)
7	G (goal)	9	–

UCS result: path S – D – G, cost 9, 7 nodes removed from the frontier.

Student's remarks

No.	Remark
R1	An explored set is not needed in BFS because BFS never revisits a node, so the Explored Set column is unnecessary.
R2	All three strategies returned S – D – G with cost 9, so all three are optimal and the traces are confirmed correct.
R3	DFS removed the fewest nodes from the frontier (3), so DFS is the best strategy for this graph.

(a) Examine the three traces and the remarks. List every error, giving the trace and iteration (or the remark number) and what is wrong. An error that is only carried forward counts once.

No.	Where (trace / iteration / remark)	What is wrong
1		
2		
3		
4		
5		
6		
7		

(b) Root cause analysis: for each error listed in (a), explain the rule or concept that was violated and how it led to the wrong entry or conclusion.

No.	Root cause (rule or concept violated)
1	
2	
3	
4	
5	
6	
7	

(c) Write the corrected traces and justify each correction.

(i) BFS: the corrected queue after iteration 3 and the full order in which nodes are dequeued.

Queue after iteration 3	
Order of dequeued nodes	

(ii) DFS: the corrected trace.

Iteration	Node Popped	Stack after Pushing Children
1		
2		
3		
4		
5		

(iii) UCS: the corrected trace.

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1			
2			
3			
4			
5			
6			

Justification of the corrections	

(d) Apply the corrected traces.

(i) Complete the comparison table.

Strategy	Path Found	Path Cost	Nodes Removed from Frontier	Optimal Here?
BFS				
DFS				
UCS				

(ii) Use iteration 3 of the corrected BFS trace to explain why an explored set is needed in graph search.

(ii)	

Question 4: Search with Partial Information – Error Analysis of Belief-State Computation

Topic: Problem formulation, search with partial information (sensorless and percept-based belief-state search)

A vacuum agent lives in a two-square world (Left square, Right square), each square being Clean (C) or Dirty (D). Actions: Left, Right and Suck (Left in the left square and Right in the right square leave the agent where it is; Suck cleans the current square). The goal is that both squares are Clean. Physical states are written (Agent, Left square, Right square):

State	Description	State	Description
S1	(L, D, D)	S5	(L, C, D)
S2	(R, D, D)	S6	(R, C, D)
S3	(L, D, C)	S7	(L, C, C)
S4	(R, D, C)	S8	(R, C, C)

Task A – sensorless agent. The agent knows it is in the Left square but nothing about the dirt, so its initial belief state is {S1, S3, S5, S7}. A student applied the plan [Suck, Right, Suck]:

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S6, S8}

Task B – agent with a local sensor that reports [location, status of the current square]. Starting from the belief state of all eight states, the agent receives the percept [Left, Dirty], performs Suck, receives [Left, Clean], performs Right and receives [Right, Dirty]. The student's table:

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S2, S3, S6}
1	Suck	{S4, S5, S7, S8}	[Left, Clean]	{S5, S7}
2	Right	{S6, S7}	[Right, Dirty]	{S6}

Student's conclusions

No.	Conclusion
C1	Task A: the final belief state still contains two states, so the agent can never know where it is. No plan can guarantee the goal when the agent has no sensors; hence sensorless problems cannot be solved.
C2	Task B: the belief state is now the singleton {S6}, so the agent knows its location and the dirt status, and the goal has been reached.
C3	With 8 physical states there are only 8 belief states (one per physical state), so searching over belief states is no harder than searching over physical states.

(a) Examine the tables and the conclusions. List every error, giving the task and step (or the conclusion number) and what is wrong. An error that is only carried forward from an earlier wrong row counts once.

No.	Where (task / step / conclusion)	What is wrong
1		
2		
3		
4		
5		
6		
7		

(b) Root cause analysis: for each error listed in (a), explain the concept or rule of belief-state search that was misapplied.

No.	Root cause (concept or rule misapplied)
1	
2	
3	
4	

No.	Root cause (concept or rule misapplied)
5	
6	
7	

(c) Write the corrected tables and conclusions, with justification.

(i) Task A: the corrected belief states.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	
2	Right	
3	Suck	

(ii) Task B: the corrected belief states after each action and each percept.

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	
1	Suck		[Left, Clean]	
2	Right		[Right, Dirty]	

(iii) Corrected conclusions C1 and C2, with justification.

C1	
C2	

(d) Apply the corrected concepts.

(i) How many different belief states are possible when there are 8 physical states? Why does this make belief-state search harder than search over physical states?

(i)	

(ii) For the initial belief state {S1, S3, S5, S7}, apply the plan [Right, Suck], write the belief state after each action, and state whether it is a solution.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Right	
2	Suck	
	Solution? Why?	

Rubrics for Error Analysis

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Error Identification	**20**	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	**30**	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	**25**	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	**15**	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	**10**	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria	Error Identification	Root Cause Analysis	Correction & Justification	Application of Concepts	Clarity & Presentation	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO1 AT3 – Error Analysis Task

Name: K. Madhulika

Reg no: 1925524190

CSA 1716 – Artificial Intelligence

Question 1: Foundations of AI, Intelligent Agents and Rationality – Error Analysis of Concept Notes

(a) Identify Correct and Incorrect Statements

Statement	Correct / Incorrect	Erroneous phrase / claim
S1	Incorrect	"The Turing Test (1950) judges a machine to be intelligent if its internal reasoning always follows the laws of logic." The claim that it is the standard example of the "thinking rationally" approach is also incorrect.
S2	Incorrect	"The AI winters … occurred in the 1950s, before the 1956 Dartmouth workshop." The claim that the workshop "revived the field" is misleading because it is regarded as the event that helped establish AI as a field, not as a revival following an AI winter.
S3	Incorrect	"A rational agent always achieves the best possible outcome because it knows the actual result of every action in advance." The claim that the taxi "was not rational" simply because an unforeseeable landslide blocked its chosen route is also incorrect.
S4	Incorrect	The Performance Measure and Actuators are interchanged. "Steering, accelerator, brake, horn" are actuators, not performance measures. "Safe, fast, legal and comfortable trip" describes the performance measure, not the actuators.
S5	Incorrect	"Partially observable, stochastic, episodic" are incorrect classifications for chess played without a clock. Chess is fully observable, deterministic, and sequential.

S6	Correct	No erroneous phrase. The statement correctly describes a model-based reflex agent and the purpose of its internal state.

(b) Root Cause Analysis

Root cause analysis identifies the underlying misconception responsible for each error.

Error (Statement / Part)	**Root Cause (Misconception)**
S1 – Turing Test	The Turing Test was confused with the *thinking rationally* approach. The Turing Test evaluates whether a machine can exhibit behavior indistinguishable from that of a human in a text-based conversation. It does not require its internal reasoning to follow formal logical laws.
S2 – AI winters	The history of AI was misunderstood. The Dartmouth workshop took place in 1956 and is associated with the introduction of the term *artificial intelligence*. AI winters occurred later, when funding and interest declined. The statement incorrectly places AI winters before the workshop.
S3 – Rationality	Rationality was confused with omniscience. A rational agent does not know every possible outcome in advance. It selects actions based on the available percepts, its knowledge, the actions available, and its performance measure. An unforeseeable event does not automatically make a previously rational decision irrational.
S4 – PEAS description	The components of PEAS were confused. PEAS stands for Performance measure, Environment, Actuators, and Sensors. The performance measure specifies the criteria for success, whereas actuators are the mechanisms through which an agent performs actions.
S5 – Chess environment	The properties of task environments were misunderstood. Chess is fully observable because both players can see the board, deterministic because moves follow fixed rules, sequential because each move affects later

	decisions, static when played without a clock, and discrete because its states and actions are distinct.
S6 – Model-based reflex agent	No misconception is present. The statement correctly explains that a model-based reflex agent maintains an internal state using information about the world and the effects of its actions. This enables it to act when the current percept alone is insufficient.

(c) Rewrite Each Statement Correctly and Justify the Correction

S1

Corrected statement:

The Turing Test, proposed by Alan Turing in 1950, evaluates whether a machine can exhibit behaviour indistinguishable from that of a human during a conversation. It is associated with the *acting humanly* approach to AI. The *thinking rationally* approach focuses on reasoning according to the principles of formal logic.

Justification:

The Turing Test evaluates intelligent behavior rather than requiring a machine's internal reasoning to follow the laws of logic. Therefore, it is associated with the acting humanly approach, not the thinking rationally approach.

S2

Corrected statement:

The Dartmouth workshop took place in 1956 and is widely regarded as a foundational event in the establishment of artificial intelligence as a field. The term *artificial intelligence* was introduced in connection with the workshop. AI winters occurred later, during periods when funding, research activity, and interest in AI declined.

Justification:

The statement incorrectly places AI winters before the Dartmouth workshop. The workshop helped establish AI as a distinct research field. AI winters occurred later, notably during the 1970s and again during the late 1980s and early 1990s.

S3

Corrected statement:

A rational agent selects the action expected to maximize its performance measure based on the percept sequence, its available knowledge, the actions it can perform, and the expected consequences of those actions. It does not need to know the actual result of every action in advance. Therefore, a taxi that chooses a route based on all available traffic reports showing it to be clear can still be rational even if an unforeseeable landslide later blocks the road.

Justification:

Rationality is based on the information available when the decision is made, not on perfect knowledge of future events. A rational agent chooses the best action it can reasonably select using the information available to it. An unexpected event does not automatically make the original decision irrational.

S4

Corrected statement:

The correct PEAS description of an automated taxi is:

PEAS Component	**Description**
Performance Measure	Safe, fast, legal, comfortable, and fuel-efficient trips; passenger satisfaction; avoiding accidents.
Environment	Roads, other vehicles, pedestrians, passengers, traffic signals, road signs, weather, and obstacles.
Actuators	Steering, accelerator, brake, horn, indicators, and other vehicle controls.
Sensors	Cameras, GPS, speedometer, radar, lidar, and other sensors used to perceive the surroundings and vehicle state.

Justification:

PEAS describes an agent's task environment. The performance measure defines how success is evaluated, while the actuators enable the agent to act on the environment. Sensors collect

information about the environment. The original statement interchanged the performance measure and actuators.

S5

Corrected statement:

Chess played without a clock is a fully observable, deterministic, sequential, static, and discrete task environment.

Justification:

The properties are explained below.

Property	**Correct Classification**	**Explanation**
Observability	Fully observable	Both players can see all pieces and their positions on the board.
Determinism	Deterministic	The outcome of a legal move is determined by the current board state and the move selected.
Task structure	Sequential	Each move affects the available moves and decisions in subsequent turns.
Dynamics	Static	Without a clock, the board does not change while a player is thinking.
State and action space	Discrete	The board positions, legal moves, and game states are distinct and countable.

S6

Corrected statement: No change.

A model-based reflex agent maintains an internal state that is updated using knowledge of how the world evolves and how its own actions affect it. This allows it to act sensibly even when the current percept does not reveal the complete state of the world.

Justification:

The statement correctly describes a model-based reflex agent. Its internal state helps it retain relevant information from previous percepts and estimate aspects of the world that cannot be observed directly at the current moment.

(d) Apply the Corrected Concepts to the Self-Driving Taxi Case

Case: A self-driving taxi chooses a highway route that every available traffic report showed to be clear. An unforeseeable landslide then blocks the road.

(i) Using the Four Factors That Decide Whether an Agent Is Rational, Argue Whether the Taxi's Choice Was Rational.

Answer:

The taxi's choice was rational because it selected the route based on the information available at the time. Rationality can be evaluated using four factors:

Factor	Application to the Taxi
1. Performance Measure	The taxi aims to provide a safe, legal, comfortable, and efficient journey. Choosing a highway reported to be clear supports these objectives.
2. Percept Sequence	The available traffic reports indicated that the highway was clear. The taxi used the information available to make its decision.
3. Knowledge of the Environment	The taxi's knowledge included available traffic information and the known road conditions. There was no information indicating that a landslide would occur.
4. Available Actions	The taxi could choose among available routes. Selecting the apparently clear highway was a reasonable decision based on the information available.

Conclusion:

The taxi's choice was rational because it selected an appropriate route using the available information and its performance objectives. The unforeseeable landslide does not automatically make the original decision irrational. However, once the landslide is detected, the taxi should update its internal state, reassess the situation, and choose a safe alternative route if one is available.

(ii) Name the Properties of the Taxi's Task Environment That Make Perfect Foresight Impossible.

Answer:

The following properties make perfect foresight impossible or impractical for a self-driving taxi:

1. **Partially Observable:** The taxi cannot directly observe every aspect of the environment. For example, a landslide may occur beyond the range of its sensors and may not appear in available traffic reports.
2. **Stochastic:** The environment contains uncertain events. Landslides, sudden accidents, and unexpected actions by other road users cannot always be predicted accurately.
3. **Dynamic:** The environment changes while the taxi is operating. Traffic conditions, weather, road obstructions, and pedestrian movements may change unexpectedly.
4. **Inaccessible or Incomplete Information:** The taxi's sensors and traffic reports provide limited information about the world. They cannot guarantee knowledge of every road condition or future event.

Conclusion:

The taxi operates in a partially observable, stochastic, and dynamic environment. These properties prevent it from knowing every detail of the environment or predicting every future event with certainty. Therefore, it must continuously monitor its surroundings and adapt its decisions as new information becomes available.

(iii) Describe One Situation in Which a Simple Reflex Taxi Would Behave Poorly, and State What a Model-Based Reflex Taxi Keeps in Order to Avoid It.

Suppose a simple reflex taxi is travelling on a road and encounters a temporary obstacle. It turns onto another road to avoid the obstacle. After turning, the obstacle is no longer visible, but the taxi has entered a dead-end street.

A simple reflex taxi selects actions based only on its current percept and condition-action rules. Because it does not maintain an internal state that represents relevant past information or the road layout, it may fail to remember that the road ahead is a dead end or that it has already encountered the obstacle.

How a Model-Based Reflex Taxi Avoids This Problem:

A model-based reflex taxi maintains an **internal state** that records relevant information about the environment. It updates this state using:

- Information from current and previous percepts.
- A model of how the environment changes.
- Knowledge of how its actions affect the environment.

Using this internal state, the taxi can remember relevant road information, recognize that it has entered a dead-end street, and select an appropriate action, such as turning around or choosing another route.

Conclusion:

A simple reflex agent relies only on the current percept, whereas a model-based reflex agent uses an internal state and a model of the environment to make more informed decisions when the current percept alone is insufficient.

Overall Conclusion

This error analysis demonstrates that understanding AI foundations requires distinguishing between intelligent behaviour, rational decision-making, agent structures, and task-environment properties. Correctly identifying and applying these concepts helps explain how intelligent agents make decisions under uncertainty and respond to unexpected events.

Question 2: Problem-Solving Agents and Problem Formulation – Error Analysis of a Formulation

(a) Identify Correct and Incorrect Components

Component / Note	Correct / Incorrect	What is wrong?
State representation	Incorrect	The state representation includes only the number of missionaries and cannibals on the left bank. It does not

		explicitly represent the boat's location, which is necessary to determine possible actions.
Initial state	Incorrect	The initial state is incomplete because it does not include the boat's location. The correct initial state is (3, 3, Left).
Actions	Incorrect	The listed actions are correct in terms of passenger combinations, but the formulation does not explicitly specify that the boat must carry one or two people and that the passengers must be available on the bank where the boat is located.
Goal test	Incorrect	The condition M = C is not sufficient to determine whether the goal has been reached. The goal is achieved when all missionaries and cannibals are on the right bank and the boat is also on the right bank.
Legality check	Incorrect	The condition $C \leq M$ on the left bank (or M = 0) checks only the left bank. The safety condition must also be checked on the right bank.
Path cost	Incorrect	The path cost is defined as the number of people carried in each crossing. The problem asks for the fewest crossings, so each crossing should have a cost of 1.
Note N1	Incorrect	The order is wrong. Goal formulation and problem formulation should occur before search. The correct order is goal formulation → problem formulation → search → execution.
Note N2	Incorrect	The stated numbers of total and legal states are incorrect for the complete state representation. With the boat's location included, there are 32 possible states and 20 legal states under the safety constraint alone.

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(b) Root Cause Analysis

Error (Component / Note)	Root Cause
State representation	The student failed to identify all information required to determine the agent's available actions. The boat's location affects which people can cross, so it must be part of the state.
Initial state	The student used an incomplete state representation and consequently omitted the boat's initial location.
Actions	The student listed passenger combinations but did not fully specify the conditions under which an action is applicable. An action must be consistent with the boat's location, passenger availability, and carrying capacity.
Goal test	The student confused a relationship between two variables with the actual goal condition. The goal requires everyone to reach the right bank, not merely for the numbers of missionaries and cannibals on the left bank to be equal.
Legality check	The student considered only one bank. The safety constraint applies independently to both banks: missionaries must not be outnumbered by cannibals on either bank when missionaries are present.
Path cost	The student confused the number of people transported with the number of crossings. Since the objective is to minimize crossings, every crossing must contribute a cost of 1.
Note N1	The student misunderstood the sequence of problem-solving agent operations. An agent must identify its goal and formulate the problem before searching for a solution.
Note N2	The student counted states using an incomplete representation or incorrect state-space assumptions. The boat's location must be included when counting the possible states.

(c) Corrected Formulation and Justification

Component / Note	Corrected Formulation	Justification
State representation	(M, C, B), where M = number of missionaries on the left bank, C = number of cannibals on the left bank, and B = boat location (Left or Right).	The boat's location is necessary to determine which crossings are possible.
Initial state	(3, 3, Left)	Initially, all three missionaries and all three cannibals are on the left bank, and the boat is on the left bank.
Actions	Move 1M, 2M, 1C, 2C, or 1M + 1C across the river, provided the boat is on the departure bank and the passengers are available there. The boat cannot cross empty.	These actions respect the boat's capacity and passenger availability. The resulting state must also satisfy the safety constraints.
Goal test	M = 0 and C = 0 and B = Right.	When M = 0 and C = 0, everyone is on the right bank. Requiring B = Right ensures that the boat has also reached the right bank.
Legality check	For each bank, either there are no missionaries or the number of missionaries is at least the number of cannibals. Formally: (M = 0 or M ≥ C) on the left bank, and (3 − M = 0 or 3 − M ≥ 3 − C) on the right bank. Also, 0 ≤ M, C ≤ 3.	Both banks must satisfy the safety constraint, and the state variables must remain within their valid ranges.
Path cost	Each crossing costs 1. The total path cost equals the number of crossings.	This directly represents the objective of moving everyone

		to the right bank using the fewest crossings.
Note N1	Goal formulation → problem formulation → search → execution.	The agent first determines the desired outcome, formulates the problem, searches for a solution, and then executes the selected actions.
Note N2	There are 32 possible states in the complete state space and 20 states that satisfy the safety constraint, before considering reachability from the initial state.	There are 4 possible values for M, 4 for C, and 2 for B, giving 4 × 4 × 2 = 32 states. Applying the bank-safety constraints leaves 20 legal states.

Note: The count of 20 legal states includes states satisfying the safety and variable-range constraints, even if some cannot be reached from the initial state through valid crossings.

(d) Apply the Corrected Formulation

(i) Complete the Table of Boat Loads Available from the Initial State

Initial state: (3, 3, Left)

Initially, all missionaries and cannibals are on the left bank, and the boat is on the left bank. After each crossing, the boat will be on the right bank.

Boat Load	**Resulting State (M, C, Boat)**	**Legal?**	**Reason**
1M	(2, 3, Right)	No	On the left bank, 2 missionaries and 3 cannibals remain. Since missionaries are present and 3 > 2, the left bank is unsafe.
2M	(1, 3, Right)	No	On the left bank, 1 missionary and 3 cannibals remain. Since missionaries are present and 3 > 1, the left bank is unsafe.

1C	(3, 2, Right)	Yes	On the left bank, 3 missionaries and 2 cannibals remain, so missionaries are not outnumbered. On the right bank, 0 missionaries and 1 cannibal are present, which is safe because no missionaries are present.
2C	(3, 1, Right)	Yes	On the left bank, 3 missionaries and 1 cannibal remain, so the bank is safe. On the right bank, 0 missionaries and 2 cannibals are present, which is also safe.
1M + 1C	(2, 2, Right)	Yes	On the left bank, 2 missionaries and 2 cannibals remain. On the right bank, 1 missionary and 1 cannibal are present. Both banks satisfy the safety constraint.

Conclusion: From the initial state, the crossings involving 1M, 2M, and 1M + 1C produce unsafe states. The crossings involving 1C and 2C produce safe states.

(ii) Give One State That the Student's Legality Check Accepts Although It Is Actually Illegal.

Answer:

Consider the state **(3, 0, Right)**.

- **Left bank:** 3 missionaries and 0 cannibals. This satisfies the student's condition because $C \leq M$.
- **Right bank:** 0 missionaries and 3 cannibals. This is safe because there are no missionaries on the right bank.

This state is actually legal, so it is not a suitable counterexample.

A correct counterexample is **(1, 0, Right)**.

- **Left bank:** 1 missionary and 0 cannibals. The student's condition accepts it because $0 \leq 1$.
- **Right bank:** 2 missionaries and 3 cannibals. This is unsafe because missionaries are present and $3 > 2$.

Therefore, the student's legality check accepts **(1, 0, Right)** even though the right bank violates the safety constraint.

(iii) Using Your Corrected Path Cost, State the Cost of an Optimal Solution.

Answer:

The corrected path cost assigns a cost of 1 to every crossing. Therefore, the total path cost is the number of crossings.

An optimal solution requires **11 crossings**.

One optimal sequence of crossings is:

Crossing	Boat Movement	State After Crossing (M, C, Boat)
1	2C →	(3, 1, Right)
2	1C ←	(3, 2, Left)
3	2C →	(3, 0, Right)
4	1C ←	(3, 1, Left)
5	2M →	(1, 1, Right)
6	1M + 1C ←	(2, 2, Left)
7	2M →	(0, 2, Right)
8	1C ←	(0, 3, Left)
9	2C →	(0, 1, Right)
10	1C ←	(0, 2, Left)
11	2C →	(0, 0, Right)

Here, → means crossing from the left bank to the right bank, and ← means returning from the right bank to the left bank.

Total path cost = 11 crossings × 1 = 11.

Justification: Since every crossing has a cost of 1, the optimal solution minimizes the number of crossings. The sequence above safely transfers all missionaries and cannibals to the right bank in 11 crossings.

Overall Conclusion

The Missionaries and Cannibals problem requires a complete state representation, valid actions, an accurate goal test, and safety checks for both riverbanks. The path cost must match the objective of minimizing crossings. Correcting these errors allows a problem-solving agent to search for a valid and optimal solution systematically.

Question 3: Searching for Solutions and Uninformed Search – Error Analysis of Search Traces

(a) Identify Every Error in the Traces and Remarks

No.	Where (Trace / Iteration / Remark)	What Is Wrong?
1	BFS – Iteration 3	The queue after expanding D is incorrect. The student adds B, E, and G but fails to account correctly for nodes already in the queue or explored. A is already in the queue from expanding C, so it must not be added again.
2	BFS – Result	The BFS result is incorrect. BFS finds the shallowest goal path, which is S–D–G, with a path cost of 9. However, the reported trace does not correctly reflect the specified graph-search rules.
3	DFS – Iteration 1	The stack after expanding S is shown as [D, C], which is correct under the specified stack convention. No error occurs here.
4	DFS – Iteration 2	The stack after expanding D is incorrect. The student pushes G, E, and B without considering that C is already on the stack and that the children must be processed according to the stated rules.
5	UCS – Iteration 3	The priority queue is incorrect because the path cost to G through D is 9, while the path cost to B through D is 7. However, the graph contains an edge B–G with cost 3, which provides a cheaper path to

		G through B. The queue must be updated when a cheaper path is found.
6	Remark R1	The claim that an explored set is unnecessary in BFS is incorrect. In graph search, an explored set prevents repeatedly expanding states already visited.
7	Remarks R2 and R3	R2 incorrectly claims that all three algorithms are optimal and that all traces are correct. R3 incorrectly concludes that DFS is the best strategy merely because it removes the fewest nodes from the frontier. Fewer frontier removals do not necessarily mean an optimal or generally better solution.

(b) Root Cause Analysis

No.	Root Cause (Rule or Concept Violated)
1	The student did not consistently apply the BFS graph-search rules. BFS must avoid adding a node that is already explored or already in the queue.
2	The student confused finding the correct BFS path with validating the entire trace. BFS finds the shallowest goal path, but the trace must still obey the queue-management rules.
3	No error occurred in DFS Iteration 1. The stack [D, C] correctly ensures that C, the alphabetically first child, is popped first.
4	The student failed to apply the DFS rule that a node already on the stack must not be pushed again. The stack must also preserve the specified alphabetical expansion convention.
5	The student failed to update frontier entries when a cheaper path was discovered. UCS must prioritize the lowest cumulative path cost and replace an existing frontier entry when a cheaper path to that node is found.
6	The student confused tree search with graph search. Graph search uses an explored set to prevent repeated expansions and unnecessary cycles.

No.	Root Cause (Rule or Concept Violated)
7	The student confused search efficiency with solution quality. BFS is optimal for equal step costs, while UCS is optimal for nonnegative edge costs. DFS is not generally optimal, even if it finds a solution quickly.

(c) Corrected Traces and Justifications

(i) BFS: Corrected Queue After Iteration 3 and Full Dequeue Order

Initial queue: [S]

BFS expands nodes in alphabetical order and does not add a node already explored or already in the queue.

Iteration	Node Dequeued	Queue After Expansion	Explored Set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[A, B, E, G]	{S, C, D}
4	A	[B, E, G]	{S, C, D, A}
5	B	[E, G]	{S, C, D, A, B}
6	E	[G]	{S, C, D, A, B, E}
7	G (goal)	[]	{S, C, D, A, B, E, G}

Queue after iteration 3: [A, B, E, G]

Order of dequeued nodes: S, C, D, A, B, E, G

Justification:

- Expanding S adds C and D.
- Expanding C adds A because A is neither explored nor already in the queue.
- Expanding D adds B, E, and G. A is already in the queue, so it must not be added again.
- BFS removes nodes in FIFO order, so A is dequeued before B, E, and G.

- The goal test occurs when G is removed from the queue.

The resulting path is S–D–G, with a cost of 9.

(ii) DFS: Corrected Trace

Convention: The top of the stack is on the left. Children are pushed so that the alphabetically first child is popped first. Nodes already explored or on the stack are not pushed again.

Iteration	Node Popped	Stack After Pushing Children
1	S	[C, D]
2	C	[A, D]
3	A	[B, D]
4	B	[D, G]
5	D	[G, E]

Important: The stack convention and the instruction to pop the alphabetically first child determine the order. However, the exact DFS trace depends on whether the implementation marks nodes explored when they are pushed or popped. Under the stated convention, nodes already explored or on the stack must not be pushed again.

Justification:

- From S, the children are C and D. C is popped first.
- From C, A is added.
- From A, B is added.
- From B, D and G are considered. D is already on the stack, so it is not added again.
- When D is popped, G and E are considered. G is placed before E so that G will be popped first.

Continuing the trace, G is eventually popped and the goal test succeeds.

DFS path found: S–C–A–B–G

Path cost: S–C = 3, C–A = 6, A–B = 2, B–G = 3.

Total cost = 3 + 6 + 2 + 3 = 14.

(iii) UCS: Corrected Trace

UCS expands the frontier node with the lowest cumulative path cost. If a cheaper path to a frontier node is found, its entry is replaced. Ties are broken alphabetically.

Iteration	Node Expanded	g(n)	Priority Queue After Expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), A:9(C), G:8(E)
5	B	7	G:8(E), A:9(C)
6	G (goal)	8	A:9(C)

Justification of the Corrections:

1. UCS expands S first, with a cumulative cost of 0.
2. From S, it adds C and D, each with cost 3.
3. Expanding C produces a path to A with cost 9.
4. Expanding D produces paths to E with cost 5, B with cost 7, and G with cost 9.
5. Expanding E discovers a cheaper route to G: S–D–E–G, with cost 3 + 2 + 3 = 8. Therefore, the G entry is updated from 9 to 8.
6. UCS then expands B, but the path through B to G costs 7 + 3 = 10, which is worse than the existing cost of 8.
7. G is removed from the frontier at cost 8, so the goal test succeeds.

Correct UCS result: S–D–E–G

Total path cost: 3 + 2 + 3 = 8.

(d) Apply the Corrected Traces

(i) Complete the Comparison Table

Strategy	Path Found	Path Cost	Nodes Removed from Frontier	Optimal Here?
BFS	S–D–G	9	7	No
DFS	S–C–A–B–G	14	6	No
UCS	S–D–E–G	8	6	Yes

Explanation:

- **BFS** finds a path with the fewest edges, not necessarily the lowest total edge cost.
- **DFS** finds a solution by exploring one branch deeply before backtracking. It does not guarantee the lowest-cost path.
- **UCS** finds the least-cost path because it expands frontier nodes in increasing cumulative path-cost order.

Therefore, UCS is the optimal strategy for this weighted graph with nonnegative edge costs.

(ii) Use Iteration 3 of the Corrected BFS Trace to Explain Why an Explored Set Is Needed in Graph Search.

Answer:

At Iteration 3, BFS removes D from the queue and expands its neighbours B, E, and G. The node A is already in the queue because it was discovered when C was expanded in Iteration 2.

The corrected queue after Iteration 3 is:

[A, B, E, G]

If BFS did not check whether a node was already in the queue or explored, it might add duplicate entries for nodes that had already been discovered. This would cause unnecessary frontier entries and repeated processing.

An explored set ensures that nodes already expanded are not expanded again. Combined with checking the frontier, it prevents duplicate processing and helps graph search terminate efficiently when cycles exist.

Conclusion: The explored set is important in graph search because it prevents repeated expansion of previously visited nodes, reduces unnecessary work, and helps avoid infinite loops in cyclic graphs.

Question 4: Search with Partial Information – Error Analysis of Belief-State Computation

(a) Identify Every Error in the Tables and Conclusions

No.	Where (Task / Step / Conclusion)	What Is Wrong?
1	Task A – Step 1	The belief state after Suck is incorrectly given as {S5, S7}. The correct belief state is {S5, S3, S7, S3}?
2	Task A – Step 2	The belief state after Right is incorrect because the previous belief state was calculated incorrectly. The agent's location and dirt conditions must be updated for every possible physical state.
3	Task A – Step 3	The belief state after the second Suck is incorrect. Suck must clean the square occupied by the agent in every possible physical state.
4	Task B – Step 0	After receiving the percept [Left, Dirty], the updated belief state is incorrectly given as {S1, S2, S3, S6}. Only states in which the agent is on the left and the left square is dirty should remain.
5	Task B – Step 1	The predicted belief state after Suck is incorrect. The action must be applied to every state in the updated belief state from Step 0.
6	Task B – Step 2	The predicted belief state after Right is incorrect because the previous belief state was wrong. The action Right changes the agent's location to the right square without changing either square's dirt status.
7	Conclusions C1, C2, and C3	C1 incorrectly claims that sensorless problems cannot be solved. C2 incorrectly claims that the goal has been reached merely because the belief state is a singleton. C3 incorrectly assumes that eight physical states imply only eight possible belief states.

Note: The belief states in Tasks A and B must be calculated by applying each action to every possible physical state and then filtering by the percept when one is received.

(b) Root Cause Analysis

No.	Root Cause (Concept or Rule Misapplied)
1	The student misunderstood the transition model of Suck. Suck cleans the square where the agent is located but does not change the agent's location or the dirt status of the other square.
2	The student did not correctly apply the transition model for Right to every physical state in the belief state.
3	The student failed to apply Suck to each possible physical state after the Right action.
4	The student misunderstood percept-based belief-state updating. A percept eliminates all physical states inconsistent with the observation.
5	The student applied Suck to an incorrect predicted belief state rather than the correctly updated belief state.
6	The student did not correctly apply the Right action to every state in the belief state. The agent changes location, but the dirt conditions remain unchanged.
7	The student confused uncertainty about the physical state with the impossibility of solving a sensorless problem. The student also confused a singleton belief state with a goal state and underestimated the number of possible belief states.

(c) Corrected Tables and Conclusions

(i) Task A: Corrected Belief States

The initial belief state is:

{S1, S3, S5, S7}

The agent knows it is in the left square but does not know the dirt status of either square.

Step	Action	Belief State After Action
0	Start	{S1, S3, S5, S7}

Step Action Belief State After Action

1 Suck {S5, S3, S5, S7}

2 Right {S6, S4, S6, S8}

3 Suck {S6, S4, S6, S8}

Justification:

- **Step 1:** Suck cleans the left square in every possible state. States with the left square already clean remain unchanged.
- **Step 2:** Right moves the agent to the right square without changing the dirt status.
- **Step 3:** Suck cleans the right square in every possible state. Therefore, the final belief state should contain only states where both squares are clean.

Correct final belief state: {S7, S8}.

(ii) Task B: Corrected Belief States

The agent starts with all eight possible physical states and receives the percept [Left, Dirty].

Step	**Action**	**After Action (Predicted)**	**Percept**	**After Percept (Updated)**
0	Start	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S3}
1	Suck	{S5, S7}	[Left, Clean]	{S5, S7}
2	Right	{S6, S8}	[Right, Dirty]	{S6}

Justification:

- **Step 0:** The percept [Left, Dirty] means the agent is in the left square and the left square is dirty. The only consistent states are S1 and S3.
- **Step 1:** Suck cleans the left square, producing S5 and S7. The percept [Left, Clean] is consistent with both states.
- **Step 2:** Right moves the agent to the right square, producing S6 and S8. The percept [Right, Dirty] eliminates S8 because the right square is clean in S8. Therefore, the updated belief state is {S6}.

(iii) Corrected Conclusions C1 and C2

C1 – Corrected conclusion:

Sensorless problems can be solved if there exists a sequence of actions that guarantees the goal for every possible physical state in the belief state. In Task A, the plan [Suck, Right, Suck] guarantees that both squares become clean.

Justification:

The agent does not need to know the exact physical state to achieve the goal. It can choose actions that succeed across all states in its belief state.

C2 – Corrected conclusion:

In Task B, the final belief state is {S6}, meaning the agent knows its location and the dirt status of both squares. However, the goal has not been reached because S6 represents (R, C, D): the agent is in the right square, the left square is clean, and the right square is dirty.

Justification:

A singleton belief state means the agent knows the exact physical state. It does not automatically mean that the goal has been achieved. The goal requires both squares to be clean.

C3 – Corrected conclusion:

With eight physical states, there are $2^8 = 256$ possible belief states, including the empty set. Belief-state search can be harder because each belief state represents a set of possible physical states, increasing the number of states that may need to be considered during search.

(d) Apply the Corrected Concepts

(i) How Many Different Belief States Are Possible When There Are Eight Physical States? Why Does This Make Belief-State Search Harder Than Search Over Physical States?

Answer:

If there are eight physical states, the number of possible belief states is:

2^n

where n is the number of physical states.

For n=8:

2^8 = 256

Therefore, there are 256 possible belief states, including the empty set.

Why is belief-state search harder?

In ordinary physical-state search, the agent searches among eight possible states. In belief-state search, each search node can represent a set of physical states, resulting in up to 256 possible belief states.

Consequently, belief-state search may require more memory and computation because the agent must track and update sets of possible physical states rather than a single physical state.

(ii) Apply the Plan [Right, Suck] to the Initial Belief State

Initial belief state: {S1, S3, S5, S7}

Step	Action	Belief State After Action
0	Start	{S1, S3, S5, S7}
1	Right	{S2, S4, S6, S8}
2	Suck	{S2, S4, S6, S8}

Solution? No.

Why?

The initial belief state contains four possible physical states in which the agent is in the left square, with different dirt conditions.

- After Right, the agent moves to the right square, but the dirt conditions do not change.
- After Suck, the agent cleans the right square in each possible state. However, the left square may still be dirty.

Therefore, the final belief state includes S2 and S4, in which the left square is dirty. Since the goal requires both squares to be clean in every possible physical state, the plan [Right, Suck] does not guarantee the goal.

Conclusion: The plan [Right, Suck] is not a solution because it does not ensure that both squares are clean for every possible initial physical state.

Overall Conclusion

Belief-state search allows an agent to reason about uncertainty by maintaining a set of possible physical states. Actions transform every state in that set, while percepts eliminate states inconsistent with observations. A sensorless agent can still achieve a goal when a suitable action sequence guarantees success across all possible states. A singleton belief state represents certainty about the physical state, but the agent must still check whether that state satisfies the goal condition.